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A Thermal Elastohydrodynamic Inlet Zone Analysis

An equation which takes account of the effects of viscous heating is used in place of the conventional Reynolds equation in a Grubin style analysis of the elastohydrodynamic inlet zone. The analysis indicates that viscous heating in the inlet zone can result in substantial reduction in the lubricant film thickness. The results of the analysis are expressed in a simple equation which will be useful for the design of elastohydrodynamically lubricated contacts under conditions where viscous heating is important.

Introduction

In the design of hydrodynamically lubricated antifriction bearings, gears and cams, the lubricant film thickness is generally calculated on the basis of isothermal elastohydrodynamic theory. The most widely used equations for the film thickness are those derived by Grubin [1]¹ and Dowson and Higginson [2]. Neither takes account of the effect of viscous heating in the inlet zone on the film formation process. Viscous heating tends to reduce the lubricant viscosity with a consequent reduction in film thickness.

The most comprehensive model of the elastohydrodynamic lubrication process has been developed by Cheng [3, 4]. His work takes account of not only the viscous heating and heat transfer within the lubricant film but also the heat transfer within the bearing surfaces. He found that viscous heating in the inlet zone could substantially reduce the film thickness [4]. However, the properties of the lubricant used in this investigation were such that the effect was only manifested at very high surface speeds.

In a recent paper, Greenwood and Kauzlarich [5] derived an approximate method of correcting for thermal effects in the inlet zone. They showed that viscous heating in the inlet zone could result in significant reductions in film thickness at all but the lowest speeds and lubricant viscosities. This finding was supported by a number of experimental investigations discussed in their paper. Their work makes it evident that it is essential to develop design equations which incorporate the effect of viscous heating in the inlet zone on the film formation process.

Wilson and Mahdavian [6] recently derived a thermal Reynolds equation which takes account of the viscosity variation across the lubricant film due to viscous heating. The use of their equation was illustrated by applying it in the analysis of a plasto-hydrodynamic inlet zone.

The use of their equation greatly facilitated the analysis. Wilson and Mahdavian used a viscosity model of the form

$$\mu = \frac{\mu_0 e^{\gamma p}}{1 + \alpha(T - T_s)} \quad (1)$$

where μ is the viscosity of the lubricant at temperature T and pressure p , μ_0 is the viscosity at zero pressure and the surface temperature T_s , γ is a pressure coefficient of viscosity and α is a temperature coefficient of viscosity. Wilson and Wong [7] used the approach of Wilson and Mahdavian to derive the thermal Reynolds equation for a more realistic viscosity model of the form

$$\mu = \mu_0 e^{\gamma p - \alpha(T - T_s)} \quad (2)$$

In this derivation they neglected the effect of slip between the surfaces since Wilson and Mahdavian had shown that viscous heating due to slip was not significant compared with that due to back flow. Wilson and Wong used their equation to analyze the film formation process in plane-strain forging.

The present work is concerned with applying the thermal Reynolds equation derived by Wilson and Wong to the elastohydrodynamic problem. This new equation is used in the place of the conventional Reynolds equation in a Grubin style analysis of the inlet zone. This yields a simple design equation which includes the effects of viscous heating in the inlet zone on the lubricant film thickness.

The system to be analyzed is shown in Fig. 1.

Assumptions Used in the Analysis

- 1 The lubricant flow is steady, incompressible and two dimensional.
- 2 Inertial effects can be neglected.
- 3 The lubricant pressure is constant across the film.
- 4 The relative slope of the surfaces is small.
- 5 The lubricant is Newtonian.
- 6 The lubricant viscosity μ is given by equation (2).
- 7 The surface temperatures are equal.
- 8 The surface velocities are equal.
- 9 The rate of viscous energy dissipation is constant across the film thickness.

¹ Numbers in brackets designate References at end of paper.

Contributed by the Lubrication Division of THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS and presented at the ASME-ASLE Joint Lubrication Conference, Montreal, Canada, October 8-10, 1974. Manuscript received by the Lubrication Division, June 19, 1974. Paper No. 74-Lub-14.

- 10 The dominant mode of heat transfer is conduction at right angles to the surfaces.
- 11 The film thickness shape in the inlet zone is that given by Hertzian theory plus a constant separation.
- 12 The inlet zone is flooded and the pressure falls to zero at a great distance from the contact zone.
- 13 The pressure reaches a high value at the edge of the contact zone.
- 14 The pressure gradient is negligible at the edge of the contact zone.

Assumptions (1) to (5) are those used in the development of the conventional Reynolds equation and require no comment here. Assumptions 6 to 10 were adopted by Wilson and Wong [7] in their analysis. The viscosity model of assumption 6 is a good approximation of the behavior of practical liquid lubricants using single pressure and temperature coefficients. Assumption 7 is accurate for many practical and almost all experimental elastohydrodynamic contacts. Wilson and Mahdavian [6] showed that sliding did not have an important effect on the film formation process in a system similar to the elastohydrodynamic inlet zone. Thus despite Assumption 8 the results of the analysis are probably valid for situations in which some sliding occurs. Assumption 9 was adopted by Wilson and Mahdavian, and Wilson and Wong to simplify their analyses. It yields similar temperature distributions to those calculated by more exact techniques and thus probably does not introduce substantial errors in the analysis. The length of elastohydrodynamic inlet zones is much larger than the film thickness and thus conduction across the film tends to be more important than conduction or convection along the film. This forms the basis for assumption 10 which does not result in any appreciable error except at exceptionally high speeds. Assumptions 11 to 14 were those adopted by Grubin [1] in his analysis of the elastohydrodynamic inlet zone.

Inlet Zone Analysis

For the system under investigation the thermal Reynolds equation derived by Wilson and Wong [7] may be written

$$F - \sqrt{\frac{\pi}{6FR}} e^{\frac{3FR}{2}} \operatorname{erf} \sqrt{\frac{3FR}{2}} + 1 = 0 \quad (3)$$

where

$$R = -\frac{p'h^3}{12\mu_s U(h-h_0)} \quad (4)$$

$$F = \frac{\mu_s \alpha U^2 (h-h_0)^2}{kh^2} \quad (5)$$

Nomenclature

$B = e^{-\gamma p}$, reduced pressure parameter

$C = \frac{H}{0.254 S^{0.73}}$, thermal correction factor

$E = \left(\frac{1-\nu_1^2}{\pi E_1} + \frac{1-\nu_2^2}{\pi E_2} \right)^{-1}$, reduced modulus

E_1, E_2 = Young's moduli of surfaces

$F = \frac{\mu_s \alpha U^2 (h-h_0)^2}{kh^2}$, thermal back-flow parameter

$H = \frac{h_0 \bar{E}}{2W}$, nondimensional film thick-

ness

$L = \frac{\mu_0 \alpha U^2}{k}$, thermal loading parameter

$R = \frac{-p'h^3}{12\mu_s U(h-h_0)}$, nondimensional pressure gradient

$S = \frac{3\mu_0 \gamma U a \bar{E}^2}{W^2}$, nondimensional speed

T = lubricant temperature

T_s = surface temperature

U = surface velocity

W = load/unit width

$X = \frac{x}{a}$, nondimensional position

a = Hertz half-width

h = film thickness

h_0 = film thickness where $\frac{dp}{dx} = 0$

k = lubricant thermal conductivity

p = lubricant pressure

$p' = \frac{dp}{dx}$, lubricant pressure gradient
 x = distance from center of contact
 α = temperature coefficient of viscosity

γ = pressure coefficient of viscosity

μ = lubricant dynamic viscosity

μ_0 = lubricant viscosity at $p = 0$ and $T = T_s$

μ_s = lubricant viscosity at $T = T_s$

ν_1, ν_2 = Poisson's ratios of surfaces

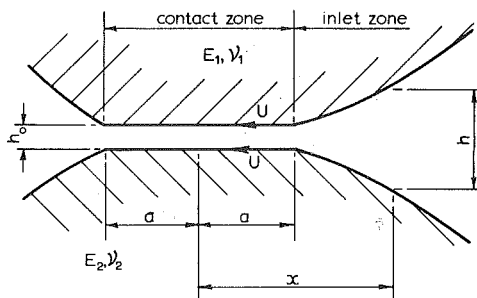


Fig. 1 System to be analyzed

h is the film thickness, h_0 is the film thickness where the pressure gradient p' vanishes, U is the surface velocity, α is the temperature coefficient of viscosity and k is the lubricant conductivity. The lubricant viscosity μ_s at pressure p and surface temperature T_s is given by

$$\mu_s = \mu_0 e^{\gamma p} \quad (6)$$

where μ_0 is the viscosity at zero pressure and temperature T_s .

Equation (3) is rather complicated for general use. Fortunately it can be approximated by

$$R = (1 + 0.608 F^{0.82})^{-1} \quad (7)$$

with sufficient accuracy for most purposes. Equation (7) will be used in the present analysis.

Equations (3) and (7) can be regarded as relationships between a nondimensional pressure gradient R and a thermal parameter F . However, in using the equations it is convenient to think of R as a correction factor which can be applied to the conventional Reynolds equation. Thus equation (4) can be written

$$\frac{dp}{dx} = -12 \frac{\mu_s U (h-h_0)}{h^3} R \quad (8)$$

which is identical to the usual Reynolds equation except for the correction factor R .

Substituting for μ_s from equation (6) in equation (8) and nondimensionalizing by writing

$$B = e^{-\gamma p} \quad (9)$$

and

$$X = \frac{x}{a} \quad (10)$$

where x is the distance from the center of contact and a is the half-width of the Hertz contact zone, yields

$$\frac{dB}{dX} = \frac{12\mu_0\gamma Ua(h - h_0)}{h^3} R \quad (11)$$

From assumptions 11 and 14 the film thickness h in the inlet zone is given by

$$h = h_0 + \frac{2W}{F} (X\sqrt{X^2 - 1} - \ln(X + \sqrt{X^2 - 1})) \quad (12)$$

where h_0 is the film thickness at the edge of the contact zone and W is the load per unit width. The reduced modulus $\bar{E}$ is given by

$$\frac{\pi}{F} = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \quad (13)$$

where E_1 and E_2 and ν_1 and ν_2 are, respectively, the Young's moduli and Poisson's ratios of the two surfaces.

Substituting for h from equation (12) into equation (11) yields

$$\frac{dB}{dX} = \frac{SYR}{(H + Y)^3} \quad (14)$$

where

$$S = \frac{3\mu_0\gamma Ua\bar{E}^2}{W^2} \quad (15)$$

$$H = \frac{h_0\bar{E}}{2W} \quad (16)$$

and

$$Y = X\sqrt{X^2 - 1} + \ln(X + \sqrt{X^2 - 1}) \quad (17)$$

From equations (5), (6), (9), (16), and (17) F is given by

$$F = \frac{Y^2 L}{B(H + Y)^2} \quad (18)$$

where L is defined by

$$L = \frac{\mu_0\alpha U^2}{k} \quad (19)$$

Substituting for F from equation (18) in equation (7) yields

$$R = [1 + 0.608(\frac{Y^2 L}{B(1 + Y)^2})^{0.82}]^{-1} \quad (20)$$

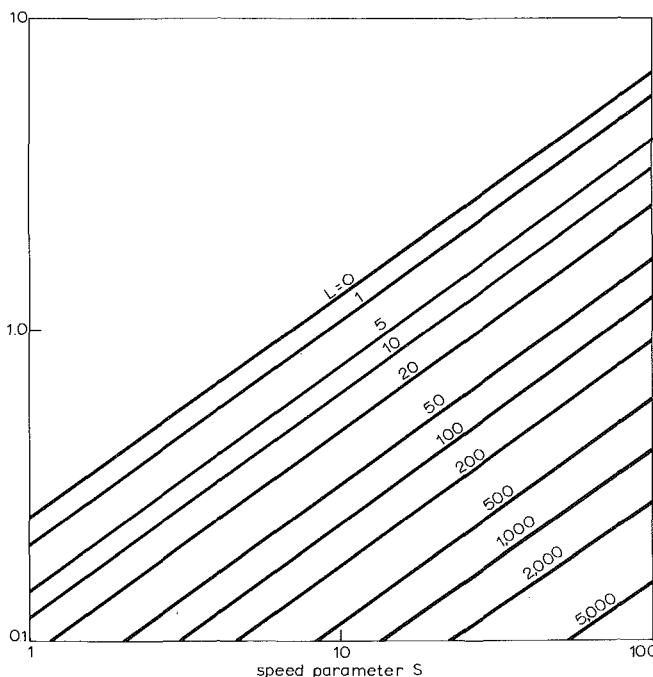


Fig. 2 Effect of thermal loading on film thickness

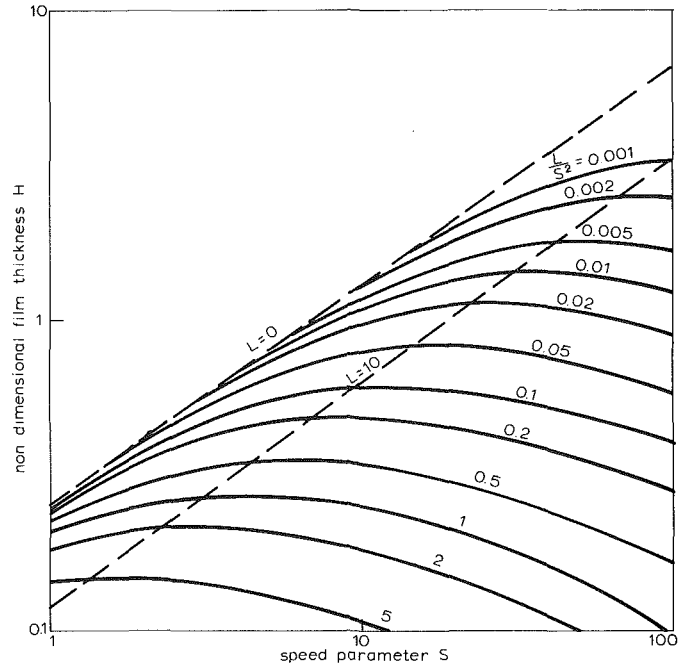


Fig. 3 Variation of film thickness with speed

Substituting for R from equation (20) in equation (14) yields

$$\frac{dB}{dX} = \frac{SY}{(H + Y)^3} [1 + 0.608(\frac{Y^2 L}{B(1 + Y)^2})^{0.82}]^{-1} \quad (21)$$

Equation (21) is subject to two boundary conditions. Assumption (12) requires that at a great distance from the contact zone when $x = \infty$, $p = 0$ or in nondimensional terms when $X = \infty$, $B = 1$. Assumption (13) requires that at the edge of contact when $x = a$, $p = \alpha$ or when $X = 1$, $B = 0$. Thus, by integrating the equation it is possible to find the relationship between the nondimensional film thickness H , the speed parameter S and the thermal loading parameter L .

A third order Runge Kutta computer program was used to integrate equation (21). Input to the program included values of S and L and an estimated value for H . Integration started at $X = 30$ (which is essentially infinite) with a value of $B = 1$ and proceeded inward to $X = 1$. If the value of B at this point was not sufficiently close to zero, the estimated value of H was adjusted and the integration repeated until the inner boundary condition was satisfied. The process was then repeated with different values of S and L . In this way, the variation of H with S and L was determined.

Results and Discussion

Fig. 2 shows the variation of the nondimensional film thickness H with the speed parameter S and the thermal loading parameter L in the range of practical interest. On a log/log plot, the curves are essentially a family of straight parallel lines. (In fact the curves are slightly concave upwards and converge as S increases.) For $L = 0$ the relationship between H and S can be written

$$H = 0.254 S^{0.73} \quad (22)$$

This relationship is equivalent to the usual isothermal Grubin equation. For values of L less than unity the film thickness is only slightly less than the isothermal value. As L is increased above unity significant progressive reduction in the film thickness occurs.

It is difficult to see the variation of film thickness with speed which is imbedded in Fig. 2. This is because both the speed pa-

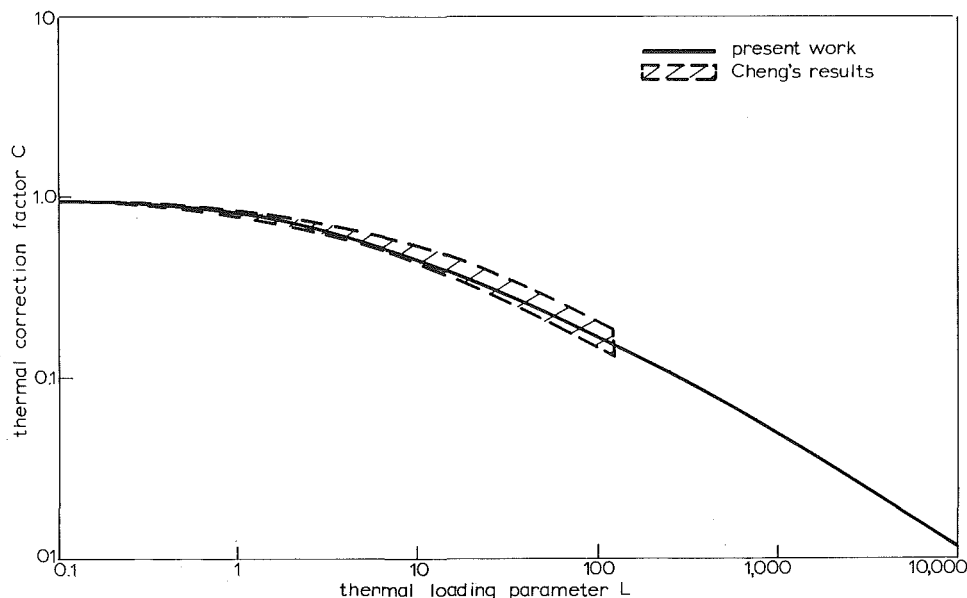


Fig. 4 Deviation of film thickness from isothermal theory

parameter S and the thermal loading parameter L change with speed. Fig. 3 replots the data shown in Fig. 2 but uses lines of constant L/S^2 rather than lines of constant L . The parameter L/S^2 is given by

$$\frac{L}{S^2} = \frac{\alpha W^4}{9\mu_0 k \gamma^2 a^2 E^4} \quad (23)$$

and is a measure of the susceptibility of the system to thermal effects which is independent of speed. Thus curves of constant L/S^2 give a true representation of the variation of film thickness with speed. From Fig. 3 it can be seen that at low speeds the film thickness follows the isothermal theory ($L = 0$). As the speed is increased thermal effects become important and the film thickness passes through a maximum and then falls off with increasing speed. The greater the value of L/S^2 the lower is the value of the speed parameter S at which the thermal effect is important. The line $L = 10$ is also plotted in Fig. 3. It can be seen that the maximum film thickness occurs in the neighborhood of this line. This is to be compared with the analyses of Greenwood and Kauzlarich [5] which predicts that the maximum film thickness will occur at values of L of 1.4 or 2.0 depending on the approximation used. It is interesting to note that the film thickness peaks predicted by the present model are relatively flat. There is a range of about an order of magnitude in speed over which the film thickness varies by less than ten percent.

It is interesting to express the results of the analysis in terms of a thermal correction factor C which is the ratio of the actual film thickness to that predicted by isothermal theory. In the present case from equation (22) C is given by

$$C = \frac{H}{0.254S^{0.73}} \quad (24)$$

Fig. 4 is a plot of C on a base of the thermal loading parameter L . Since the curves in Fig. 2 are essentially parallel a single curve has been plotted. For values of L less than unity, C has a value near unity. For higher values of L there is a marked decrease in C with increase in L . The relationship between C and L can be represented approximately by

$$C = \frac{3.94}{3.94 + L^{0.62}} \quad (25)$$

Substituting for C from equation (25) into equation (24) and

rearranging yields

$$H = S^{0.73}(3.94 + L^{0.62})^{-1} \quad (26)$$

Equation (26) can be used as a means of predicting the film thickness in an elastohydrodynamic contact which takes into account the effect of viscous heating in the inlet zone. For small values of L it reduces to Grubin's equation [1]. It is also possible to derive a similar equation based on Dowson and Higginson's work by applying the correction factor C to their equation.

The range of the results of Cheng quoted by Dyson [8] is also shown in Fig. 4. Since Cheng used a different viscosity model to that used in the present work, the comparison has been based on matching the rates of change of viscosity with temperature as suggested by Dyson. It is evident that the correction factors indicated by the present work are in close agreement with those calculated by Cheng. The differences can probably be explained by the different viscosity model and the more exact heat transfer model which Cheng used. However, the close agreement provides support for the various simplifying assumptions used in the present work.

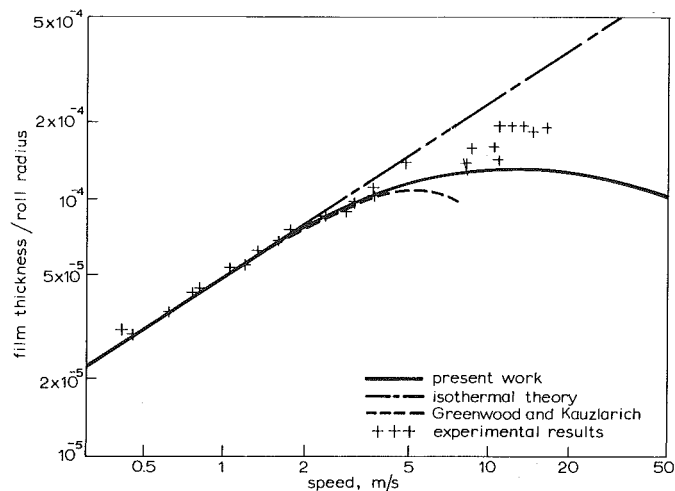


Fig. 5 Comparison of theories with experiment

Fig. 5 compares the deviation from isothermal theory predicted by the present work with the experimental results of Dyson, Naylor and Wilson [9] (oil A at 40 C). The isothermal line for this data given by Greenwood and Kauzlarich was assumed to be correct. It is evident that the present work is in better agreement with these results than the Greenwood and Kauzlarich model. The improvement is most marked at the higher speeds where the Greenwood and Kauzlarich model substantially underestimates the film thickness. However, the experimental data does not extend to high enough speeds to support the fall off in film thickness predicted by the present theory.

It is interesting to speculate on the reasons for the difference between Greenwood and Kauzlarich's model and the present work. The authors believe that Greenwood and Kauzlarich's assumption that the pressure distribution in the inlet zone is unaffected by viscous heating is largely responsible. In the present work it was found that as the thermal loading parameter was increased the pressure gradients in the inlet zone decreased. This resulted in the effective length of the inlet zone increasing from about $5a$ under isothermal conditions to around $30a$ under conditions of high thermal loading. This effect would explain why Greenwood and Kauzlarich's model yields smaller films than the present work. The increased length of the inlet zone could also mean that lubricant starvation effects may become more important under conditions of high thermal loading.

Conclusion

Wilson and Wong's thermal Reynolds equation has been used in place of the conventional Reynolds equation in a Grubin style analysis of the film formation process in elastohydrodynamic lubrication. This allows the effect of viscous heating in the inlet zone on the lubricant film thickness to be taken into account. The results can be conveniently expressed in the form

$$H = S^{0.73}(3.94 + L^{0.62})^{-1} \quad (26)$$

where H is the nondimensional film thickness, S a speed parameter and L is a thermal loading parameter. Equation (26) appears to be a useful tool for the design of highly loaded line contacts.

Acknowledgment

The authors would like to thank the University of Massachusetts Computer Center for the use of their facilities.

References

- 1 Grubin, A. N., and Vinogradova, I. E., "Investigation of the Contact of Machine Components," Ts NITMA Sh, Book 30, Moscow, 1949, (D.S.I.R., London, Translation No. 337).
- 2 Dowson, D., and Higginson, G. R., "A Numerical Solution for the Elastohydrodynamic Problem," *Journal of Mechanical Engineering Science*, Vol. 1., 1959, pp. 6-15.
- 3 Cheng, H. S., "A Refined Solution to the Thermal E.H.L. of Rolling and Sliding Cylinders," *Trans ASLE*, Vol. 8, 1965, pp. 397-410.
- 4 Cheng, H. S., "Isothermal Elastohydrodynamic Lubrication Theory in the Full Range of Pressure Viscosity Coefficient," *TRANS. ASME, Journal of Lubrication Technology*, Vol. 94, Ser. F, No. 1, 1972, pp. 35-43.
- 5 Greenwood, J. A., and Kauzlarich, J. J., "Inlet Shear Heating in Elastohydrodynamic Lubrication," *TRANS. ASME, Journal of Lubrication Technology*, Vol. 95, Ser. F, No. 4, 1973, pp. 417-426.
- 6 Wilson, W. R. D., and Mahdavian, S. M., "A Thermal Reynolds Equation and Its Application to the Analysis of Elastohydrodynamic Inlet Zones," *JOURNAL OF LUBRICATION TECHNOLOGY*, *TRANS. ASME*, Series F, Vol. 96, No. 4, Oct. 1974, pp. 572-577.
- 7 Wilson, W. R. D., and Wong, C. J., "Analysis of the Film Formation Process in Plane Strain Forging," *JOURNAL OF LUBRICATION TECHNOLOGY*, *TRANS. ASME*, Series F, Vol. 96, No. 4, Oct. 1974, pp. 605-610.
- 8 Dyson, A., discussion of reference [5].
- 9 Dyson, A., Naylor, H., and Wilson, A. R., "The Measurement of Oil Film Thickness in Elastohydrodynamic Contacts," *Proceedings of the Institute of Mechanical Engineers*, Vol. 180, Pt. 3B, 1966, pp. 119-134.